

LABORATORY DIGITAL PROCESSING SIGNAL

2022-2023

INSTRUCTIONS

The assignments must fully comply with the instructions, otherwise they will not be graded. The relevant instructions are formulated as follows:

The execution and writing of the work is **individual**, as is the delivery. Collaborations are not permitted.

All answers will be collected in ONE doc file. This file will contain:

• The student's full details

• All the **pronunciations** of the exercises in the order they have been given. Below the pronunciation of the exercise to be answered, the solution will follow. In the event that an exercise is not to be answered, there will be only its pronunciation.

• The solution to each exercise will **necessarily include**:

- o The code (in Matlab, Octave ... software) that solves – answers the question.
- o Explanation of the commands used
- o Commenting on the solution
- o The m files you will create (copy –paste)
- o Commenting on the results
- o Theoretical solution (where necessary)
- o Figures, graphs related to the solution and results of the Exercise

This file will be delivered - without any compression - to a special directory in eclass, at the end of the laboratory classes.

The assignments will be processed for plagiarism checking by academic paper detection, copying and plagiarism software. In case the same answer is found in 2 or more assignments (even in one of the questions), all these assignments will be nullified.

Some assignments may also be examined orally.

Good luck!

EXERCISES FOR DELIVERY

1. Draw the discrete-time signal

$$x[n] = \begin{cases} 2, & -4 \leq n \leq 4 \\ 4, & 4 < n \leq 10 \\ 10, & 10 < n \leq 20 \end{cases}$$

Where AM is the student's registration number () the single-digit sum of and its digits.

2. Design the discrete-time signals

$$a) y[n] = u(n - (3 + AM \bmod 5)) - 6 - (n - AM \bmod 4)$$

$$b) x[n] = (n + 2 + AM \bmod 5) - (n - 2 - AM \bmod 4)$$

for $n = -50 - (AM \bmod 4) : 50 + (AM \bmod 4)$

Where AM is the student registration number and $u[n]$ and $\delta[n]$ are the unitary step and unitary impulse sequence respectively.

3. Draw the following signals and for those that are periodic, determine their fundamental period and verify the periodicity graphically.

$$a) y_1[n] = 8 \cos\left(\frac{11}{6}n\right)$$

$$b) x_2[n] = 2 \left(\frac{1}{3} + \frac{1}{4} + 0.4 \right)$$

$$c) x_3[n] = \left(\frac{1}{100} \right) (0.25)^n$$

$$d) x_4[n] = 3 \left(\frac{1}{6} + \frac{1}{10} \right)^2$$

$$e) x_5[n] = \frac{1}{2} + \frac{1}{2} \delta[n]$$

4. Write a function that accepts as an argument a discrete-time signal and the time interval in which it is defined and returns the even and odd parts of the signal, as well as the time axis on which they are defined. Finally, the function should create a graph of the two output signals, on the correct time axis, e.g. $[x_e, x_o, m] = \text{ev_od}(x, n)$. Then, using the function, calculate the even and odd parts of the sequence $x[n] = [\text{the digits of AM}, 1, \text{the digits of AM}]$, where the unit in the center corresponds to the time point $n=0$. Make the graphs.

(for example, for AM= 17098, $x[n] = [1\ 7\ 0\ 9\ 8\ 1\ 1\ 7\ 0\ 9\ 8]$)

5. Calculate the output of the LTI system:

a) with impulse response $\tilde{y}[n] = \delta[n] - 3\delta[n-1] + 6\delta[n-2]$ and input

$$\begin{aligned} x_1[n] &= 0 \\ x_2[n] &= 1 \\ x[n] &= \{x_1[n], x_2[n]\} \end{aligned}$$

Setting $n_1 = -10$ to $n_2 = 10$.

b) with impulse response $\tilde{y}[n] = 0.6\delta[n]$ and input $x[n] = \delta[n] - \delta[n-10]$

Setting $n_1 = -10$ to $n_2 = 30$.

6. Draw the convolution of the sequences $x[n] * y[n]$ where

$$x[n] = \delta[n-4] - \delta[n-5]$$

$$y[n] = (1 - \delta[n-5]) - (\delta[n] - \delta[n-5])$$

The convolution of the two sequences should be written in the form of a Toeplitz vector matrix product by a vector. Write a function that implements the convolution of two sequences by calculating the convolution as a Toeplitz vector matrix product by a vector.

7. To investigate graphically the linearity of the system described by the relationship:

$$y[n] = 3x[n] + 4.$$

The following signals should be considered as inputs:

$$x_1[n] = [1\ 2\ 3\ 4\ 5], \quad x_2[n] = \delta[n] - \delta[n-4], \quad x_3[n] = \delta[n] - \delta[n-6].$$

Consider $\ddot{y} = 2$ and $\ddot{y} = 3$ as constants and investigate in the interval $n = [-5 : 10]$. Draw the inputs x_1 , x_2 and the two outputs in subwindows of the same graphic window.

8. The difference equation of a system is given:

$$[y] + 3[y(1)] + 5[y(2)] = 0.5[y] + [y(2)]$$

a) Calculate and graph the impact response.

b) From this graph, explain whether the system is BIBO stable or not.

c) Calculate the output for input the sequence $x[n] = [2 \ 1 \ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \ 2]$ using the filter function and the conv function and explain what you see in relation to what we commented on in the brochure.

d) Repeat question (c) for input $x[n] = 2\cos(3\ddot{y} n)$.

9. Given a system described by the ED

$$[y] = 0.4[y(1)] + [y] + 0.7[y(2)]$$

a) Find and plot the output for time $n = [-20 : 20]$ when the input it is

$$[y] = 2[y(1)] + 4[y] + 8[y(1)] + 9[y(2)]$$

b) Find and plot the impulse response of the system for $n = [-20 : 20]$ using the impz function (help impz for syntax).

10. The signal is given.

$$[y] = 2\ddot{y} \cos\left(\frac{\ddot{y}}{8}\right), n=0:60.$$

To this signal, Gaussian noise with a standard deviation of 0.5 is added and the noisy signal is input to the system described by the difference equation:

$$[] = \frac{1}{\ddot{y}} \ddot{y} [\ddot{y}]$$

This system is a mean value filter of order M. Write a program that calculates the output of the system for M=4. Plot the quiet signal, the noisy signal, and the output of the system. What do you think is the action of the system on the noisy signal?

11. Design the step responses of the systems described by the following input-output relationships:

$$a) y(n) = x(n+2) + x(n-2)$$

$$b) y(n) = x^2(n-2)$$

12. Find the sequences that have the following Z-transformations, and then calculate their first 6 terms

$$1(z) = \frac{1}{z^2}$$

$$2(z) = \frac{3z^2}{(z^2 - 1.5z + 0.5)(z - 0.5)}$$

$$3(z) = \frac{z^2 + 7}{z^2 + z - 2}$$

13. Analyze the rational function into a sum of simple fractions:

$$F(z) = \frac{z^2 + 3z - 1}{z^3 - 5z^2 + 8z - 4}$$

14. Given an LTI system with impulse response $h(n)=0.9^n u(n)$. Find its transfer function, investigate whether it is stable, and draw the pole-zero diagram.

15. Analyze the rational function into a sum of simple fractions:

$$H(z) = \frac{z^4 - 1.1z^3 + 0.88z^2 - 2.396z + 1.348}{z^3 - 0.7z^2 - 0.14z + 0.048}$$

16. Find the Transfer Function of the system described by the Difference Equation

$$y(n) + 1.5y(n-1) + 0.5y(n-2) = x(n) + x(n-1)$$

in two ways a) using *tf()* b) using *solve()*. To check stability, to make a pole-zero diagram.

17. Calculate the transfer function and the impulse response of the causal first-order discrete-time FFA system, which is described by the difference equation:

$$y(n) - 0.9y(n-1) = x(n)$$

Draw the pole-zero diagram and determine the PS.

18. Given the causal discrete-time MFA system whose input and output are connected by the difference equation

$$y(n) - 0.5y(n-1) = x(n)$$

Calculate the output of the system if the input signal is $x(n) = u(n)$ and the system is at rest.

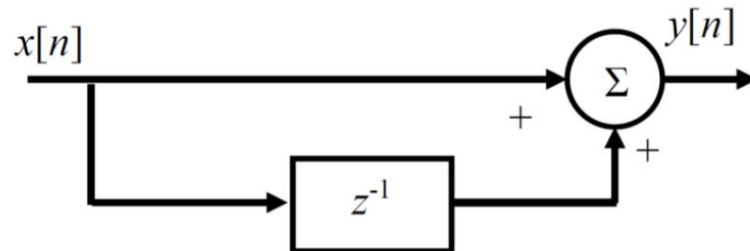
19. A discrete GFA system is described by the difference equation:

$$[] = [] \ddot{y} [\ddot{y} 1] + [\ddot{y} 2] .$$

Draw: a) the impulse response and b) the gain and phase diagrams as a function of frequency.

Is a calculation of the impact response required?

20. Plot the magnitude and phase as a function of the frequency of the MF. discrete-time system described in the figure.



21. In the above system, input $x[n] = u[n]$ is applied . Design the output $y[n]$ for $0 \leq n \leq 200$. How is the system characterized in terms of frequency pass?

22. . Given the sequence

$$() = 0.7 \quad , \quad 0 \leq n \leq 19$$

Plot the measure of the DTFT for $0 \leq n \leq 20$ and the DFT over the frequencies in the same figure:

$$\ddot{y} = 2\ddot{y}k/N, k = 0, \dots, N-1.$$

23. Write a function that implements the linear convolution of two sequences using the functions `fft()`, `ifft()`.

24. Calculate the linear convolution of the sequences using the above function:

$$[] = [1 \ 2 \ 3 \ 4], 0 \leq n \leq 3,$$

$$[] = [7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1], 0 \leq n \leq 6.$$

Confirm the result using `conv()`.

25. The following sequences are given:

$$[x] = [1, 2, 3, 4, 5] \quad [y] = [1, 1, 2, 1], \quad 0 \leq n \leq 3$$

Calculate their circular convolution a) in the time domain b) using `fft()`, `ifft()`.

26. For the two sequences of the previous exercise:

- Calculate the 8-point circular convolution
- Find the linear convolution a) in the time domain b) using `fft()`, `ifft()`

27. Calculate the power of the unit step sequence:

$$u[n], \text{ for } -5000 \leq n \leq 5000.$$

Compare the result you found with the result from the theoretical solution.

28. Draw the magnitude, phase, real part, imaginary part of the DTFT of the sequence

$$[x] = (0.5)^n, \quad 0 \leq n \leq 30$$

in $N=501$ equally spaced frequencies between 0 and π .
Why is this observation interval (0 to π) sufficient?

Note, the DTFT calculation should be done based on the definition. In the DTFT measure diagram, the measure based on the theoretical value should be captured (hold on) as well (use the formula from the theory) as well as the DTFT measure calculated by `freqz()`.

29. Draw the magnitude, phase, real part, and imaginary part of the DTFT of the sequence

$$[x] = (0.9)^n \exp(j\pi n/3), \quad 0 \leq n \leq 10$$

at $N=501$ equidistant frequencies. What observation interval is sufficient?

30. Determine the linearity property for the DTFT with the following two discrete-time signals:

$$x_1[n] = (1, 11), \quad x_2[n] = (1, 11)$$

and $n=2, n=3$, in the interval $0 \leq n \leq 10$.